

Shayan Shakeri

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PROFESSIONAL SUMMARY

AI Research Engineer at BMO's AI Centre of Excellence. Built BMO Wealth's multi-agent data layer from the ground up, scaled it, and own it today: an orchestration harness giving agents graph, analysis, and search tools over client data. Built the eval pipeline that caught a production GenAI tool serving \$200B+ AUM downplaying investment risk. Took a proprietary subcutaneous neurostimulator from idea to production device at Epineuron. Co-founder of Merlyn Labs; open-sourced a flow-matching VLA integration for RLinf. Canadian citizen, eligible to work in the UK without sponsorship via the Youth Mobility Scheme or High Potential Individual visa.

CORE COMPETENCIES

Agentic Systems

Agent Evaluation

LLM Fine-tuning

Reinforcement Learning

Python & C/C++

PyTorch & JAX

WORK EXPERIENCE

BMO AI Centre of Excellence Sep 2025 - Present

AI Research Engineer

Toronto, Canada

- Built, scaled, and own BMO Wealth's multi-agent data layer: an orchestration harness with graph, analysis, and search tools.
- One text request: screened 30,000 clients for defined-benefit plans autonomously in 3 hours, down from a work week.
- Built a deterministic eval pipeline that caught a production GenAI tool serving \$200B+ AUM downplaying investment risk.
- Designing evals for LLM agents that retrieve and reason over commercial banking and insurance policy.

Merlyn Labs Aug 2025 - Present

AI Researcher & Co-Founder

Toronto, Canada

- Open-sourced a flow-matching VLA integration for RLinf, enabling RL training on the BEHAVIOR-1K suite in OmniGibson.
- Developing VLM judges that score rollouts into dense, context-dependent RL rewards that are difficult to game.
- Showed $\pi 0.5$'s LIBERO-PRO failures are recipe-induced overfitting; proposed a recalibrated baseline for the benchmark.
- Conservative finetuning doubled position-swap success (21% to 42%) with no architectural change to the model.

Epineuron Technologies May 2021 - Aug 2022

Biomedical & Electrical Engineering Co-op

Mississauga, Canada

- Owned a proprietary subcutaneous neurostimulator from first idea to production device at a clinical-stage startup.
- Designed and assembled the PCBs and wrote the IEC 60601-1 / ISO 13485 validation protocols for clinical test runs.
- Part of the team behind PeriPulse, an FDA Breakthrough-designated device now in multinational clinical trials.
- Extended battery life 900% through oscilloscope and power-analyzer testing of the device's power draw.
- Modeled electromagnetic field penetration into peripheral nerves in COMSOL; the results set the electrode diameter.

PROJECTS

Stanford BEHAVIOR-1K Challenge 8th Place, Standard Track

Identified proprioceptive collapse as a critical VLA failure mode (60% masking improved task success up to 48%); found chunked execution outperformed temporal ensembling 3x. Trained on 10,000+ demonstrations covering 22 of 50 scored task types; published a technical report and a LessWrong analysis.

BardSong Closed Alpha

Built the end-to-end pipeline: Groq speech-to-text, Gemini scene extraction, image generation, narrated video. Fine-tuned an LLM on D&D sourcebooks for narrative coherence; in closed alpha with 23 dungeon masters.

EDUCATION

M.Eng in AI & Robotics University of Toronto Sep 2024 - Apr 2027 (anticipated)

B.Eng, Engineering Physics Co-op McMaster University Sep 2018 - Jun 2023

Dean's Honour List. Coursework: Computational Multiphysics, Quantum Computing, Numerical Methods, Signals & Systems.

SKILLS

Agents & ML: Agents, Agent Evaluation, Alignment Evaluation, LLM Fine-tuning, Reinforcement Learning, Imitation Learning, Flow Matching
Systems & Languages: Python, C/C++, PyTorch, JAX, OmniGibson, MuJoCo, ROS2, Sim-to-Real, PCB Design